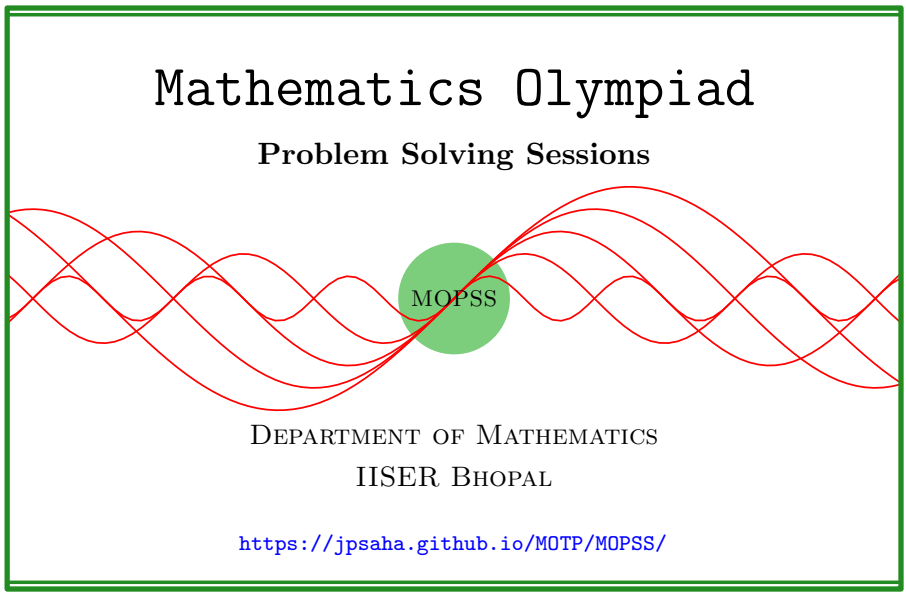
Cyclic quadrilaterals

MOPSS



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

List of problems and examples

1 Exercise (RMO 1992 P5,) . . . 2
2 Exercise (RMO 1992 P4,) . . . 2
3 Exercise (RMO 1999 P3,) . . . 3
4 Exercise (RMO 2000 P5,) . . . 3
5 Exercise (RMO 2001 P1,) . . . 3
6 Exercise (RMO 2002 P5,) . . . 3
7 Exercise (RMO 2006 P1,) . . . 3
8 Exercise (RMO 2008 P1,) . . . 3
9 Exercise (RMO 2010 P5,) . . . 3
10 Exercise (RMO 2011a P5,) . . . 3
11 Exercise (RMO 2011a P1,) . . . 3
12 Exercise (RMO 2011b P1,) . . . 4
13 Exercise (RMO 2013a P1,) . . . 4
14 Exercise (RMO 2013c P3,) . . . 4
15 Exercise (RMO 2014c P1,) . . . 4
16 Exercise (RMO 2014c P5,) . . . 4
17 Exercise (RMO 2014a P5,) . . . 4
18 Exercise (RMO 2014e P6,) . . . 4
19 Exercise (RMO 2014d P5,) . . . 4
20 Exercise (RMO 2015c P5,) . . . 4
21 Exercise (RMO 2015d P5,) . . . 4
22 Exercise (RMO 2015b P1,) . . . 5
23 Exercise (RMO 2015d P1,) . . . 5
24 Exercise (RMO 2015a P5,) . . . 5
25 Exercise (RMO 2015e P1,) . . . 5

§1 Cyclic Quadrilaterals

Exercise 1 (RMO 1992 P5,). $ABCD$ is a cyclic quadrilateral, x, y, z are the distances of A from the lines BD, BC, CD respectively. Prove that

$$\frac{BD}{x} = \frac{BC}{y} + \frac{CD}{z}.$$

Exercise 2 (RMO 1992 P4,). $ABCD$ is a cyclic quadrilateral with $AC \perp BD$; AC meets BD at E. Prove that

$$EA^2 + EB^2 + EC^2 + ED^2 = 4r^2,$$

where r is the radius of the circumscribing circle.

Exercise 3 (RMO 1999 P3,). Let $ABCD$ be a square and M, N points on sides AB, BC, respectively, such that $\angle MDN = 45°$. If R is the midpoint of MN show that $RP = RQ$ where P, Q are the points of intersection of AC with the lines MD, ND.

Exercise 4 (RMO 2000 P5,). The internal bisector of angle A in a triangle ABC with $AC > AB$, meets the circumcircle Γ of the triangle in D. Join D to the centre O of the circle Γ and suppose DO meets AC in E, possibly when extended. Given that BE is perpendicular to AD, show that AO is parallel to BD.

Exercise 5 (RMO 2001 P1,). Let BE and CF be the altitudes of an acute triangle ABC, with E on AC and F on AB. Let O be the point of intersection of BE and CF. Take any line KL through O with K on AB and L on AC. Suppose M and N are located on BE and CF respectively, such that KM is perpendicular to BE and LN is perpendicular to CF. Prove that FM is parallel to EN.

Exercise 6 (RMO 2002 P5,). The circumference of a circle is divided into eight arcs by a convex quadrilateral $ABCD$, with four arcs lying inside the quadrilateral and the remaining four lying outside it. The lengths of the arcs lying inside the quadrilateral are denoted by p, q, r, s in counter-clockwise direction starting from some arc. Suppose $p + r = q + s$. Prove that $ABCD$ is a cyclic quadrilateral.

Exercise 7 (RMO 2006 P1,). Let ABC be an acute-angled triangle and let D, E, F be the feet of perpendiculars from A, B, C respectively to BC, CA, AB. Let the perpendiculars from F to CB, CA, AD, BE meet them in P, Q, M, N respectively. Prove that P, Q, M, N are collinear.

Exercise 8 (RMO 2008 P1,). Let ABC be an acute angled triangle; let D, F be the midpoints of BC, AB respectively. Let the perpendicular from F to AC and the perpendicular from B to BC meet in N. Prove that ND is the circumradius of ABC.

Exercise 9 (RMO 2010 P5,). Let ABC be a triangle in which $\angle A = 60°$. Let BE and CF be the bisectors of $\angle B$ and $\angle C$ with E on AC and F on AB. Let M be the reflection of A in line EF. Prove that M lies on BC.

Exercise 10 (RMO 2011a P5,). Let ABC be a triangle and let BB_1, CC_1 be respectively the bisectors of $\angle B, \angle C$ with B_1 on AC and C_1 on AB. Let E, F be the feet of perpendiculars drawn from A onto BB_1, CC_1 respectively. Suppose D is the point at which the incircle of ABC touches AB. Prove that $AD = EF$.

Exercise 11 (RMO 2011a P1,). Let ABC be a triangle. Let D, E, F be points respectively on the segments BC, CA, AB such that AD, BE, CF concur at the point K. Suppose $\frac{BD}{DC} = \frac{BF}{FA}$ and $\angle ADB = \angle AFC$. Prove that $\angle ABE = \angle CAD$.

Exercise 12 (RMO 2011b P1,). Let ABC be an acute angled scalene triangle with circumcentre O and orthocentre H. If M is the midpoint of BC, then show that AO and HM intersect on the circumcircle of ABC.

Exercise 13 (RMO 2013a P1,). Let ABC be an acute-angled triangle. The circle Γ with BC as diameter intersects AB and AC again at P and Q, respectively. Determine $\angle BAC$ given that the orthocenter of triangle APQ lies on Γ.

Exercise 14 (RMO 2013c P3,). In an acute-angled triangle ABC with $AB < AC$, the circle ω touches AB at B and passes through C intersecting AC again at D. Prove that the orthocentre of triangle ABD lies on ω if and only if it lies on the perpendicular bisector of BC.

Exercise 15 (RMO 2014c P1,). Let ABC be a triangle with $\angle ABC$ as the largest angle. Let R be its circumcenter. Let the circumcircle of triangle ARB cut AC again at X. Prove that RX is perpendicular to BC.

Exercise 16 (RMO 2014c P5,). Let ABC be a triangle and let X be on BC such that $AX = AB$. let AX meet circumcircle ω of triangle ABC again at D. Prove that circumcentre of triangle BDX lies on ω.

Exercise 17 (RMO 2014a P5,). Let ABC be an acute angled triangle with H as its orthocentre. For any point P on the circumcircle of triangle ABC, let Q be the point of intersection of the line BH with line AP. Show that there is a unique point X on the circumcircle of triangle ABC such that for every P other than B, C, the circumcircle of HPQ passes through X.

Exercise 18 (RMO 2014e P6,). Let D, E, F be the points of contact of the incircle of an acute-angled triangle ABC with BC, CA, AB respectively. Let I_1, I_2, I_3 be the incentres of the triangles AFE, BDF, CED, respectively. Prove that the lines I_1D, I_2E, I_3F are concurrent.

Exercise 19 (RMO 2014d P5,). Let ABC be an acute-angled triangle and let I be its incentre. Let the incircle of triangle ABC touch BC in D. The incircle of the triangle ABD touches AB in E; the incircle of the triangle ACD touches BC in F. Prove that B, E, I, F are concyclic.

Exercise 20 (RMO 2015c P5,). Let ABC be a right angled triangle with $\angle B = 90^\circ$ and let BD be the altitude from B on to AC. Draw $DE \perp AB$ and $DF \perp BC$. Let P, Q, R and S be respectively the incenters of $\triangle DFC, \triangle DBF, \triangle DEB$ and $\triangle DAE$. Suppose S, R, Q are collinear. Prove that P, Q, R, D are concyclic.

Exercise 21 (RMO 2015d P5,). Two circles Γ and Σ in the plane intersect at two distinct points A and B, and the center of Σ lies on Γ. Let points C and D be on Γ and Σ respectively, such that C, B, D are collinear. Let point E on Σ be such that $DE \parallel AC$. Show that $AE = AB$.

Exercise 22 (RMO 2015b P1,). Let ABC be a triangle. Let B' denote the reflection of B in the internal angle bisector ℓ of $\angle A$. Show that the circumcenter of $\triangle CB'I$ lies on the line ℓ, where I is the incenter of $\triangle ABC$.

Exercise 23 (RMO 2015d P1,). Let ABC be a triangle. Let B', C' denote respectively the reflection of B, C in the internal angle bisector of $\angle A$. Show that the triangles ABC and $AB'C'$ have the same incenter.

Exercise 24 (RMO 2015a P5,). Let ABC be a triangle with circumcircle Γ and incenter I. Let the internal angle bisectors of $\angle A, \angle B, \angle C$ meet Γ in A', B', C' respectively. Let $B'C'$ intersect AA' at P, and AC in Q. Let BB' intersect AC in R. Suppose the quadrilateral $PIRQ$ is a kite, *i.e.*, $IP = IR$ and $QP = QR$. Prove that ABC is an equilateral triangle.

Exercise 25 (RMO 2015e P1,). In a cyclic quadrilateral $ABCD$, let the diagonals AC and BD intersect at X. Let the circumcircles of triangles AXD and BXC intersect again at Y. If X is the incenter of triangle ABY, show that $\angle CAD = 90°$.

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)